

Last Time:

- CMGs

Today:

- Sensor Noise
 - Power Spectral Density
 - Gauss-Markov Process
 - Allan Variance / Deviation
 - Data Sheets
-

A Closer Look at Sensor Noise:

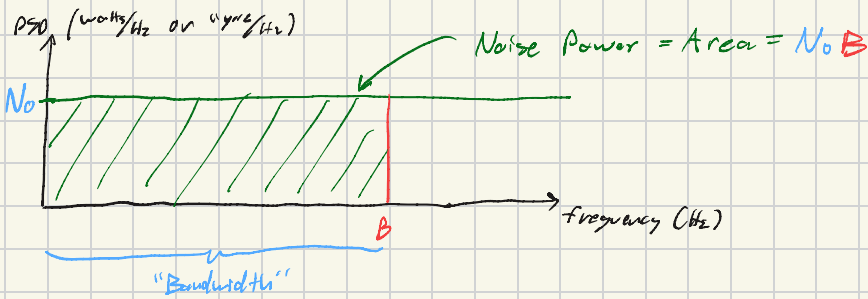
- So far, we've mostly assumed $AWGN$ ^{← Gaussian} ^{← Noise}
_{additive} _{white}
- This is standard in many fields, is mathematically convenient, and is a good description of many real-life scenarios.
- In discrete time, this is modeled as:

$$y_n = g(x_n) + w_n, \quad w_n \sim N(0, W)$$

_{measurement} _{measurement model} _{Noise Sample} _{zero mean} _{Covariance}

- Note: we can assume mean $E[w_n] = 0$ because we can include the mean in $g(x)$.
- In the frequency domain, we can look at the noise power spectral density (PSD)

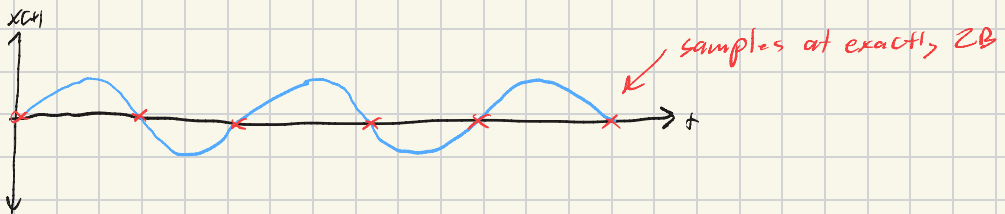
- For AWGN, the PSD is flat:



- For discrete-time signals, $B \sim$ sample rate (Nyquist sampling theorem).

Nyquist + Review:

- To reconstruct a continuous-time signal $x(t)$, we must sample at a rate greater than $2B$ ($f_s > 2B$)
- Easy to see for a pure sine wave:



- The result extends to general $x(t)$ by decomposing into a sum of sine waves (Fourier transform)
- Any frequency content $> 2B$ causes "aliasing" in sampled signal.
- In practice, we have to low-pass filter $x(t)$ before sampling to make sure there is very little power at frequencies $> 2B$

- AWGN doesn't really make sense in continuous time ($B \rightarrow \infty \Rightarrow \text{noise power} \rightarrow \infty$)
- The covariance we sample w_n from is directly related to noise PSD:

$$W = \underbrace{E[w_n^2]}_{(\text{signal})^2 \sim \text{"power"}} \sim N_0 B$$

- If the underlying (analog) signal is band-limited (with $B < 2f_s$), we want to keep noise power constant while changing f_s in our filter/sim:

$$\Rightarrow W \sim \frac{N_0 B}{f_s} \sim N_0 B \Delta t \quad \text{ } \underbrace{\Delta t}_{\text{sampling time}}$$

- Noise PSD is typically quoted as $\sqrt{N_0}$.

$$\Rightarrow w/\text{Hz} \sim (\text{signal})^2/\text{Hz} \rightarrow (\text{signal})/\sqrt{\text{Hz}}$$

- For a gyro this will have units of eg. $\text{deg/sec}/\sqrt{\text{Hz}}$
- To turn this into variance, square and multiply by bandwidth
- Bandwidth can be tricky to figure out as it is controlled by internal low-pass filtering, but is often not configurable or well documented.

Beyond AWGN - Random Walk:

- If we sit a gyro on the bench and collect data, it will have a non-zero mean (bias) in addition to AWGN
- The bias "wanders" over long time scales
- A simple idealized model of this "wandering bias" is a random walk:

$$b_{n+1} = b_n + v_n, \quad v_n \sim N(0, V)$$

- This is an integral or running sum of a white-noise process.
- The expected distance from the origin grows with time:

$$E[\|b_n\|] = \sqrt{E[b_n^2]} = \sqrt{\sum_{i=1}^n E[v_i^2]} = \sqrt{nV}$$

$\Rightarrow$ Covariance grows like n (or t) while distance $= \sqrt{\text{cov}(b_n)}$ goes like $\sqrt{n}$ (or $\sqrt{t}$).

- Confusingly, AWGN for gyros is usually called "angle random walk" on data sheets because AWGN as we integrates to a random walk on σ
- The drift on the bias that we're modeling as a random walk is called "bias instability"
- Random walk is only an approximation of the true noise statistics on $b(t)$ chain for convenience for the EKF. You can do better, but there are diminishing returns.

- Our overall noise model on y_n for a gyro combines AWGN with a random walk:

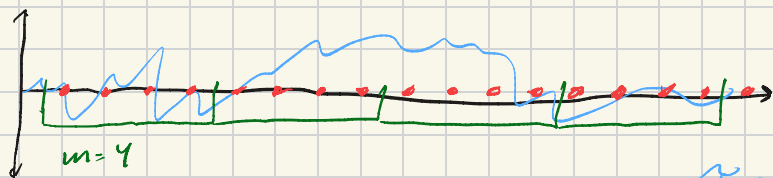
$$y_n = g(x_n) \cdot b_n + w_n, \quad w_n \sim N(0, W)$$

$$b_{n+1} = b_n + v_n, \quad v_n \sim N(0, V)$$

- This is called a "first-order Gauss-Markov Process" because it's a 1D linear system (integrator) driven by white noise.
- Can go to higher order, but then you have more filter states and can have observability issues.
- Nearly all practical filters use the 1st order model.

Allan Variance:

- How do we measure W and V ?
- Collect data from a stationary gyro $y_{i:n}$
- Can't just measure $\text{cov}(y_{i:n})$ because of wandering mean (variance will not converge due to random walk).
- Allan Variance solves this problem by looking at how mean changes over time as a function of averaging time τ
- For a discrete-time signal $y_{i:n}$ at sample rate τ_0 :



- Split data into bins of m samples. Compute average of each bin:

$$\bar{X}_n = \frac{1}{m} \sum_{k=m(n-1)+1}^{mn} y_k$$

- Allan variance is variance of the difference between adjacent bins:

$$A(\tau) = \frac{1}{2} E[(\bar{X}_{n+1} - \bar{X}_n)^2]$$

↑
averaging
time

↑
factor of $1/2$ by convention

- Allen deviation is $\sqrt{A}$ by analogy with variance $\rightarrow$ std. dev.
- Typically plot $\sigma = \sqrt{A(\tau)}$ vs. τ on a log-log plot

⊛ Intuition:

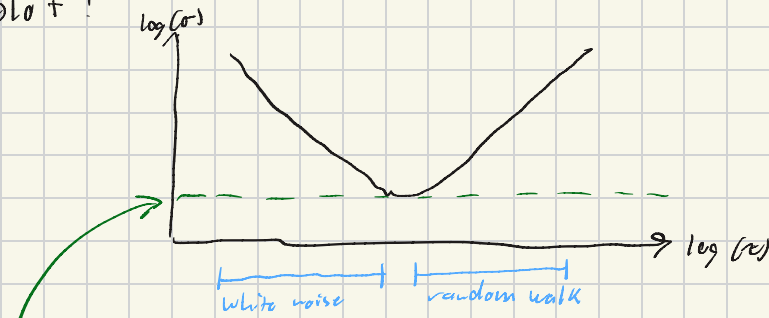
- For a pure AWGN signal, the longer τ is, the closer we get to converging to zero (like $1/\sqrt{\tau}$)

$\Rightarrow \sigma$ vs τ has downward slope ($-1/2$ on log-log plot)

- For a pure random walk, the longer τ is, the bigger the distance we wander (like $\sqrt{\tau}$)

$\Rightarrow \sigma$ vs τ has upward slope ($+1/2$ on log-log plot)

- If we sum these we get a V-shaped Allan deviation plot!



This minimum is what's typically quoted in data sheets as "bias instability"